

Alloying behavior, microstructure and mechanical properties in a FeNiCrCo_{0.3}Al_{0.7} high entropy alloy



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ABSTRACT

The present paper reports the synthesis of FeNiCrCo_{0.3}Al_{0.7} high entropy alloy (HEA) by mechanical alloying (MA) and spark plasma sintering (SPS) process. Alloying behavior, microstructure, mechanical properties and detailed phases of the alloy were investigated systematically. During MA, the formation of a supersaturated solid solution with body-centered cubic (BCC) structure occurred. However, partial BCC structure phase transformed into a face-center cubic (FCC) structure phase during SPS. Two BCC phases with nearly the same lattice parameter of 3.01 Å and one FCC phase with the lattice parameter of 3.72 Å were characterized in the transmission electron microscope (TEM) images. The two BCC phases which are evidently deviated from the definition of high entropy alloys (HEAs) are enriched in Fe–Cr and enriched in Ni–Al, respectively. Moreover, the FCC phase agrees well with the definition of HEAs. Bulk FeNiCrCo_{0.3}Al_{0.7} alloy with little porosity exhibits much better mechanical properties except compression ratio compared with other typical HEAs of FeNiCrCoAl HEA system. The yield strength, compressive strength, compression ratio and Vickers hardness of FeNiCrCo_{0.3}Al_{0.7} alloy are 2033 ± 41 MPa, 2635 ± 55 MPa, 8.12 ± 0.51% and 624 ± 26 H_v, respectively. The fracture mechanism of bulk FeNiCrCo_{0.3}Al_{0.7} alloy is dominated by intercrystalline fracture and quasi-cleavage fracture.

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1. Introduction

In order to expand the scope of engineering and industrial applications, a wide range of alloy systems with traditional design notions which are based on one or two dominant elements have been exploited, e.g. Mg-based alloys [1–3], bulk metallic glass alloys (BMGs) [4,5] and intermetallics [6]. According to conventional metallurgical theory, alloy systems with multi-principal elements can easily lead to the formation of intermetallic compounds or other complicated compounds which exhibit complex microstructure and poor mechanical properties. However, high entropy alloys (HEAs) which have been developed by Yeh et al. have broken-through this thinking pattern since 2004 [7]. Generally, HEAs contain at least five dominant elements, the concentration of each element is between 5 at.% and 35 at.%. It is of interest that HEAs exhibit simple solid solutions with BCC and/or FCC structure(s) and excellent properties due to high entropy mixing. The relevant investigations reveal that HEAs display excellent anticorrosive properties [8], high hardness [9], high thermal stability [10,11], superior compressive properties [12], excellent wear resistance [13,14], excellent high-temperature softening resistance and unique magnetic properties [7,15], etc. These desirable properties

facilitate HEAs possessing promising future for engineering and industrial applications.

Although previous researches have shown that many HEA systems have been exploited and synthesized by arc-melting (casting), the phases of HEAs are just characterized by crystal structure (FCC and/or BCC structures), demonstrating that definite phases of HEAs have been rarely investigated. Moreover, only a few of HEAs processed by mechanical alloying have been reported [16–20]. Existing literatures show that CuFeNiCrCoAl HEA system and FeNiCrCoAl HEA system prepared by arc-melting has been extensively studied, especially CuFeNiCrCoAl HEA system. In this work, we concentrated on FeNiCrCoAl HEA system. Wang et al. investigated microstructure and compressive properties of the as-cast FeNiCrCoAl HEA deeply [21]. Shun and Du has studied microstructure and tensile behaviors of the as-cast FeNiCrCoAl_{0.3} HEA after 700 °C and 900 °C isothermal aging [22]. Moreover, microstructure, hardness, corrosion properties, electrical properties, magnetic properties and Hall properties of FeNiCrCoAl_x (0 ≤ x ≤ 2) HEAs fabricated by arc-melting have been investigated systematically [23–26]. Apparently, all of the reported alloys of FeNiCrCoAl HEA system are prepared by arc-melting (casting), meanwhile, detailed phases of FeNiCrCoAl HEA system have not been deeply investigated. The lack of investigations about the detailed phases has hindered the establishment of structure–property relationship in FeNiCrCoAl HEA system. In addition, bulk HEAs with high performance can be achieved via MA and SPS which can rapidly consolidate the HEA powders to high density

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within short soaking time owing to rapid heating and cooling rates [27,28]. On the other hand, the addition of Co element can soften HEAs of FeNiCrCoAl_{0.5} HEA system while the addition of Al element can strengthen HEAs of FeNiCrCoAl_{0.5} HEA system [29]. Thus, the concentrations of Co and Al in HEAs of FeNiCrCoAl HEA system should be in an appropriate range with the aim to achieve excellent mechanical properties. Hence, in the present paper, FeNiCrCo_{0.3}Al_{0.7} HEA was designed and successfully synthesized by MA and SPS. Alloying behavior, microstructure, detailed phases and mechanical properties of FeNiCrCo_{0.3}Al_{0.7} HEA were intensively investigated.

2. Experimental procedures

FeNiCrCo_{0.3}Al_{0.7} (in atomic proportion) HEA was prepared by 45 h dry milling of mixed elemental powders. Elemental powders Fe, Ni, Cr, Co and Al with high purity (>99.9 wt.%) and particle size of $\leq 45 \mu\text{m}$ (325 mesh) were processed by mechanical alloying. The mechanical alloying process was carried out in a high performance stainless steel vial into which the powders were sealed together with tungsten carbide balls (5 mm in diameter) under an Ar atmosphere. The sealed vial was then fitted in a high energy planetary ball milling machine (QM-3SP2 Planetary Ball Mill) at 300 rpm. The ball to powder weight ratio was 10:1. Ball milled powder was then consolidated by SPS (Dr. Sinter 825, Sumitomo Coal Mining Co. Ltd., Japan) in a 20.4 mm-inner-diameter graphite die at 1000 °C for 8 min. Sintering was performed under vacuum (residual cell pressure < 8 Pa) with a constant compression pressure of 30 MPa. For easy removal and significant reduction of temperature inhomogeneities, 0.2 mm thick graphite foils were placed between the punches and the powders, and between the die and the powders, respectively. For all the sintering, the samples were heated to 600 °C within 4 min. From 600 °C to 900 °C and from 900 °C to 1000 °C, heating rates of 75 °C min⁻¹ and 50 °C min⁻¹ were used, respectively.

A Bruker D8 ADVANCE X-ray diffractometer (XRD) with a Cu K α radiation was used for the analysis of the milled powders and phase evolution after sintering. Microstructural analysis was employed by a Zeiss Supra 40 (Carl Zeiss NTS GmbH, Germany) scanning electron microscope (SEM). For the microstructure observation, the samples were polished and then etched with aqua regia solution. Thin-foil specimens were prepared by mechanical thinning followed by ion milling at room temperature and were observed under a transmission electron microscope (200 kV TEM, JEM-2100, JEOL, Tokyo, Japan) with selected area electron diffraction (SAED) analysis. The thermal analysis was carried out in a NETZSCH STA 449C differential scanning calorimeter (DSC) at a heating rate of 10 K/min under flowing high purity argon atmosphere. According to GB/T 7314-2005 [30], the ambient temperature mechanical properties were evaluated by uniaxial compression tests on samples of $\varnothing 3 \text{ mm} \times 4.5 \text{ mm}$ by using an Instron 5500 testing system with a strain rate of $1 \times 10^{-3} \text{ s}^{-1}$. Three compression tests were performed to obtain average value. Hardness measurement was carried out on the consolidated samples at a load of 300 gf in Digital Micro Hardness Tester HVS-1000 Vickers hardness instrument. The reported hardness value was an average of at least 10 measurements.

3. Results and discussion

3.1. Alloying behavior

Fig. 1 shows the XRD patterns of FeNiCrCo_{0.3}Al_{0.7} HEA powder with different milling time. Diffraction patterns of all the pure elements can be predominantly observed in the initiate time (0 h). The peaks of both Al and Co disappear simultaneously with drastic

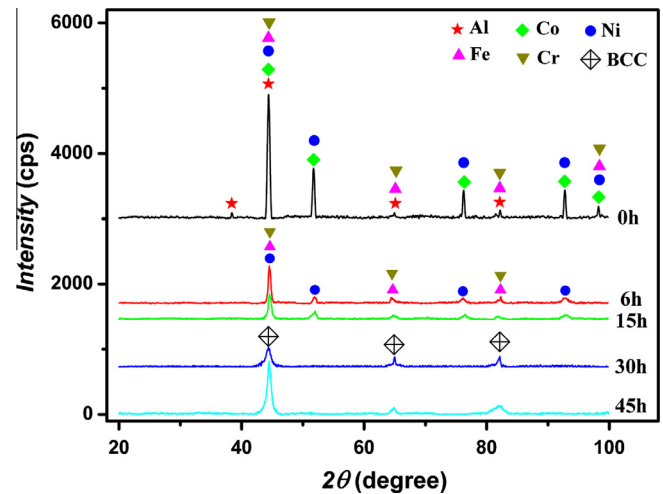


Fig. 1. XRD patterns of FeNiCrCo_{0.3}Al_{0.7} HEA powder with different milling time.

decrement of diffraction intensity as the milling time increases to 6 h. As the milling time extends to 15 h, the peaks are almost the same as the peaks of the 6 h ball milled powder except for a minor decrement of diffraction intensity. Both Al and Co dissolve most rapidly which is associated with lower melting point compared with other elements [31]. On the other hand, these two elements possess smaller atomic fractions, which also can lead to dissolve rapidly. Most of the elemental peaks essentially disappear within 30 h of milling, indicating that the formation of complete solid solution occurs. It is noted that a solid solution with BCC crystal structure has been achieved within 30 h of milling. Subsequently, the milling time was extended to 45 h with the aim to obtain refined grains and homogeneous solid solution. The diffraction peaks exhibit no evident change as the milling time reaches up to 45 h.

The crystallite size and lattice strain calculated by Scherrer's formula of the BCC phase of 30 h and 45 h of milling are shown in Table 1. The calculated results in Table 1 represent that the crystallite size of the BCC phase is greatly refined as the milling time increases to 45 h. Meanwhile, the lattice strain of 45 h of milling is predominantly higher than that of 30 h of milling. Apparently, the circulation of crushment and agglomeration during MA result in the crystallite refinement. The increment of lattice strain is associated with size mismatch effect between the constituents, increasing grain boundary fraction caused by crystallite refinement and increasing dislocation density produced by severe plastic deformation during MA [19,28]. Finally, all of other elements appear to be dissolving into Fe or Cr peaks as the milling time increases during the MA process. It is possibly explained that Fe and Cr possess more open BCC crystal structure, thus they may accommodate other elements without much expansion [17]. Moreover, Cr has the highest melting point compared with other elements leading to dissolve most slowly, which also can account for the above-mentioned phenomenon [31].

In FeNiCrCoAl_x ($0 \leq x \leq 2$) HEA system, HEAs possess single FCC phase when $x = 0$, 0.25, and 0.375, while displaying BCC and FCC duplex structures when $x = 0.50$ and 0.75, however, the crystal

Table 1
The crystalline size and lattice strain of FeNiCrCo_{0.3}Al_{0.7} HEA during MA.

Crystal structure	Milling time (h)	Crystalline size (nm)	Lattice strain (%)
BCC	30	21	0.35
BCC	45	12	0.67

structure from $x = 0.875$ to $x = 2.00$ presents single BCC phase [23]. In other words, the concentration of Al is 15.79 at.% when $x = 0.75$ and is 17.95 at.% when $x = 0.875$ in FeNiCrCoAl_x ($0 \leq x \leq 2$) HEA system. However, the concentration of Al is 17.5 at.% in FeNiCrCo_{0.3}Al_{0.7} HEA. The decrement of the concentration of Co element equals the increment of concentration of Al element. Obviously, when the concentration of Al is between 15.79 at.% ($x = 0.75$) and 17.95 at.% ($x = 0.875$), the crystal structure of HEAs in FeNiCrCoAl_x ($0 \leq x \leq 2$) HEA system is undiscovered. In FeNiCrCo_{0.3}Al_{0.7} HEA, the concentration of Al is 17.5 at.% which is close to 17.95 at.% when the alloy possesses only BCC structure in FeNiCrCoAl_x ($0 \leq x \leq 2$) HEA system. FeNiCrCo_{0.3}Al_{0.7} HEA might display a single BCC structure. Fig. 1 proves that the solid solution indeed exhibits only BCC crystal structure after 30 h of milling. The solid solution might be a supersaturated solid solution [19]. The formation of this supersaturated solid solution is attributed to the energy stored in the grain boundaries, high entropy of mixing and solid solubility extension [18,32].

In addition, according to the calculation of most of reported HEAs, Yang et al. proposed that solid solutions with simple crystal structures in HEAs are formed when $A \geq 1.1$ and $\delta \leq 6.6\%$ [33]. The parameter A and δ are defined as the following two equations:

$$A = \frac{T_m \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|} \quad (1)$$

where $T_m = \sum_{i=1}^n C_i (T_m)_i$ and $(T_m)_i$ is the melting point of the i th component of alloy. $\Delta H_{\text{mix}} = \sum_{i=1}^n \sum_{j=1}^n \Omega_{ij} C_i C_j$. Here Ω_{ij} ($= \Delta H_{ij}^{\text{mix}}$) is the regular solution interaction parameter between the i th and j th elements, C_i or C_j is the atomic percentage of the i th or j th component, and $\Delta H_{ij}^{\text{mix}}$ is the enthalpy of mixing of binary liquid alloys [34]. In FeNiCrCo_{0.3}Al_{0.7} HEA system, the chemical mixing enthalpy of binary systems is shown in Table 2. $\Delta S_{\text{mix}} = -R \sum_{i=1}^n (C_i \ln C_i)$. Where C_i is mole percent of component, $\sum_{i=1}^n C_i = 1$ and R ($= 8.314 \text{ kJ}^{-1} \text{ mol}^{-1}$) is the gas constant.

$$\delta = \sqrt{\sum_{i=1}^n C_i (1 - r_i/\bar{r})^2} \quad (2)$$

Here C_i is mole percent of component, r_i is the atomic radius of i th component, and $\bar{r}$ is the mean atomic radius of all elements.

The calculated results of A and δ for FeNiCrCo_{0.3}Al_{0.7} HEA are 1.96 and 5.48%, implying that simple BCC structured solid solution can be theoretically formed in FeNiCrCo_{0.3}Al_{0.7} HEA during MA.

3.2. Phase analysis and microstructure after SPS

Fig. 2 illustrates the XRD pattern of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS at 1000 °C for 8 min. The XRD pattern of FeNiCrCo_{0.3}Al_{0.7} HEA consolidated by SPS reveals that both FCC phase and BCC phase are visible. Based on the XRD pattern result, the BCC phase is most likely to be the main phase of the alloy after SPS, and the calculated lattice parameters are 3.0096 Å (BCC) and 3.7195 Å (FCC), respectively. As compared to only one BCC phase of 30 h and 45 h of milling during MA, partial BCC phase evolved into an FCC phase during the SPS process. In general, MA is a non-equilibrium process facil-

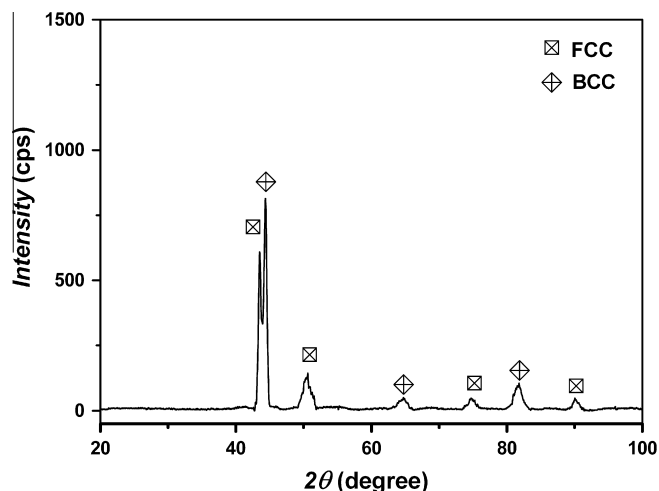


Fig. 2. XRD pattern of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS.

itates the formation of metastable supersaturated solid solution [17,19]. At high temperature (1000 °C), reordering has resulted in more stable phases during the SPS process. Reordering can be caused by the annihilation of defects introduced into the structure due to high-energy ball milling during the MA process. As a result, defects introduced by severe plastic deformation during MA are nearly annihilated after consolidation by SPS, which can also contribute to phase evolution during SPS [17]. Furthermore, the short-time exposure sintering at 1000 °C is an extremely complicated process, and the huge pulsed electric current during the SPS process might also lead to uncertain phase evolution [27]. In summary, presence of the FCC phase implies that densification at 1000 °C for 8 min has resulted in more stable phases.

The SEM micrographs of FeNiCrCo_{0.3}Al_{0.7} HEA consolidated by SPS are shown in Fig. 3. In Fig. 3a, very little porosity is observed in the consolidated FeNiCrCo_{0.3}Al_{0.7} HEA. The relative density calculated with respect to actual density and theoretical density is 99.18%. Low-magnified image (Fig. 3a) shows two distinctive areas, viz., irregularly bulk areas and small irregular areas. Three distinctive areas are evidently observed in high-magnified image (Fig. 3b), viz., irregularly bulk areas (Region 1) and two kinds of small irregular areas (Regions 2 and 3). It is obvious that the phases of Region 3 have been etched away by aqua regia solution. It means that there are at least three phases in FeNiCrCo_{0.3}Al_{0.7} HEA. Most of the areas are not more than 1 μm except the irregularly bulk areas, demonstrating short-time sintering during the SPS process cannot supply enough time for grains to grow.

Fig. 4 illustrates the TEM bright field image and corresponding SAED patterns of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. Fig. 4a shows bright field TEM image of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. However, all of the phases cannot be easily identified by contrast in this bright field image. Chemical compositions (in at.%) of the phases of bulk FeNiCrCo_{0.3}Al_{0.7} HEA are detected by EDS/TEM, and crystal structures are determined by the corresponding SAED patterns of the phases. The chemical composition analysis results are listed in Table 3, indicating that there are three different phases. According to the corresponding SAED patterns of these three phases, they are classified as two BCC phases and one FCC phase. In order to differentiate these three phases, they are named BCC1, BCC2 and FCC hereinafter. Most of the regions of bright field image are labeled in Fig. 4a with the aim to distinguish all of the phases. In addition, the two BCC phases are the main phases based on the XRD pattern result in Fig. 2. According to the chemical composition analysis results in Table 3, it is obvious that the BCC1 phase is enriched in Fe–Cr, the BCC2 phase is enriched in Ni–Al, while the FCC phase

Table 2

The chemical mixing enthalpy ($\Delta H_{ij}^{\text{mix}}$, kJ/mol) of binary equiatomic alloys calculated by Miedema's approach [34].

Element (atomic sizes, Å)	Co	Fe	Ni	Cr	Al
Co (1.24)	–	–1	0	–4	–19
Fe (1.24)	–	–	–2	–1	–11
Ni (1.25)	–	–	–	–7	–22
Cr (1.25)	–	–	–	–	–10
Al (1.43)	–	–	–	–	–

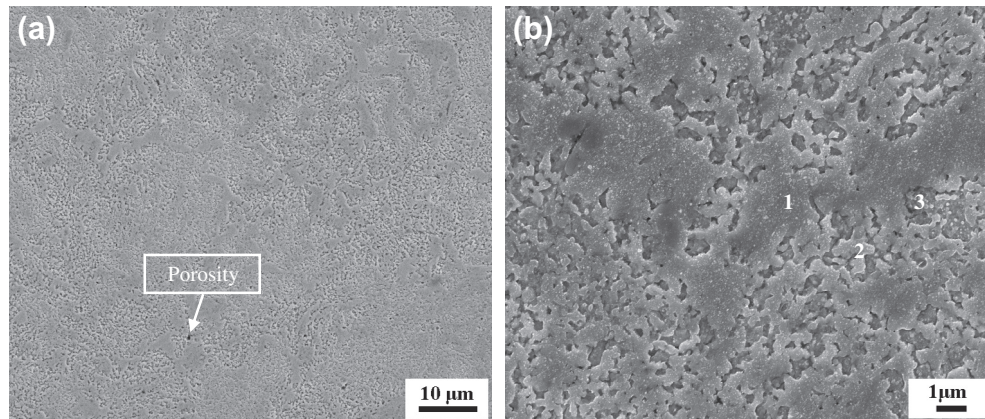


Fig. 3. SEM micrographs of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. (a) Low-magnified image and (b) high-magnified image.

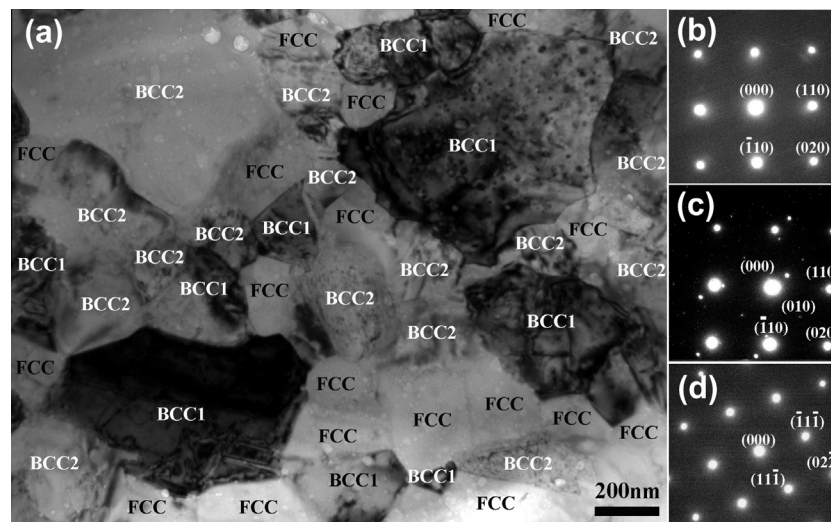


Fig. 4. TEM image and corresponding SAED patterns of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. (a) Bright field image, (b) SAED pattern of BCC1 [001] zone axis, (c) SAED pattern of BCC2 [001] zone axis, showing weak superlattice reflections corresponding to B2 structure and (d) SAED pattern of FCC [011] zone axis.

Table 3

Chemical composition (in at.%) analysis results of the phases by EDS/TEM.

Phases	Crystal structure	Fe	Ni	Cr	Co	Al
Nominal composition	–	25	25	25	7.5	17.5
BCC1 phase	BCC (A2)	30.8 ± 2	13.3 ± 2	43.5 ± 2	6.6 ± 2	5.8 ± 2
BCC2 phase	BCC (B2)	13.6 ± 2	43.7 ± 2	8.4 ± 2	8.2 ± 2	26.1 ± 2
FCC phase	FCC	28.6 ± 2	23.6 ± 2	28.3 ± 2	11.6 ± 2	7.9 ± 2

is enriched in Fe–Cr–Co. The BCC1 phase with dark contrast displays the largest grain size of these phases, exhibiting approximately 800 nm in the thickest direction. The grain size of the BCC2 phase shows a wide distribution ranging from several tens of nanometers to several hundred nanometers. The FCC phase is slightly finer than these two BCC phases, which agrees well with that the FCC phase is transformed from the initial BCC phase of the MA process.

Interestingly, the XRD pattern of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS characterized only one BCC crystal structure and one FCC crystal structure (Fig. 2). The [001] zone axis diffraction pattern (Fig. 4b) displays the lattice reflections generated by the BCC1 phase of a disordered BCC structure (A2). The lattice parameter of the BCC1 phase as measured by SAED is 3.01 Å. Meanwhile, the superlattice reflections corresponding to an ordered BCC struc-

ture (B2) are also visible in Fig. 4c. Moreover, the lattice parameter of the BCC2 phase measured by SAED is 3.01 Å, and equals to that of the BCC1 phase. It means that these two BCC phases possess nearly the same crystal structure, but their chemical compositions are on different levels. The lattice parameter of the FCC phase measured by SAED is 3.72 Å (Fig. 4d). Thus, only one BCC crystal structure and one FCC crystal structure were characterized in the XRD pattern of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. It is worth pointing out that the lattice parameters of the three phases measured by SAED are pretty close to the results calculated by the XRD pattern.

Based on the chemical compositions of these three phases, however, the BCC1 phase (Fe–Cr-rich phase) with A2 structure and the BCC2 phase (Ni–Al-rich phase) with B2 structure are evidently deviated from the definition of HEAs. The Fe–Cr-rich phase and Al–Ni-rich phase of AlCrFeNiCoCu HEA have been reported in

Ref. [35,36]. In addition, in the present paper, the FCC phase fits the definition of HEAs well. According to chemical mixing enthalpy of a binary Ni–Al system (as shown in Table 2), the largest negative enthalpy of mixing which is -22 kJ/mol results in the formation of the Ni–Al-rich phase [34,35]. The Ni–Al-rich phase is also slightly enriched in Co while depleted Fe and Cr, it can be explained that Co has high solubility in this type Ni–Al-rich phase [36]. Wang et al. regarded this type Ni–Al-rich phase as NiAl intermetallics because of the superstructure [36], however, the lattice parameter of this B2 phase is 3.01 Å, which does not equals to that of NiAl intermetallics. Therefore, this Ni–Al-rich phase of FeNiCrCo_{0.3}Al_{0.7} HEA might not be NiAl intermetallics. According to previous investigations, Cr and Co prefer the Ni sites, whereas Fe prefers the Al sites [35,37]. As a result, it could be postulated that the stoichiometry of the Ni–Al-rich phase of bulk FeNiCrCo_{0.3}Al_{0.7} HEA could be written as $(\text{Ni}_x\text{Co}_y\text{Cr}_z)_{-}(\text{Al}_i\text{Fe}_w)_+$, Ni and Al are the dominant compositions.

Although the enthalpy of mixing of a Cr–Fe binary system is only -1 kJ/mol (Table 2), Fe and Cr jointly segregate into Fe–Cr-rich phase, the formation of this phase is much more complicated [35]. The Fe–Cr-rich phase with A2 structure, exhibiting a lattice parameter of 3.01 Å. The Fe–Cr-rich phase of AlCrFeNiCoCu HEA has been reported as $(\alpha\text{-Fe, Cr})$ solid solution in Ref. [36]. However, the Fe–Cr-rich phase of FeNiCrCo_{0.3}Al_{0.7} HEA is greatly enriched in Fe–Cr. Thus, by considering the lattice parameter and the phase chemical composition, the Fe–Cr-rich phase could not be assumed as $(\alpha\text{-Fe, Cr})$ solid solution. Hence, this Fe–Cr-rich phase of FeNiCrCo_{0.3}Al_{0.7} HEA also could be written as $(\text{Ni}_x\text{Co}_y\text{Cr}_z)_{-}(\text{Al}_i\text{Fe}_w)_+$, whereas Cr and Fe are the main compositions. The FCC phase agrees well with the definition of HEAs. Due to high mixing entropy of HEA systems, stable phases are expected to be formed in HEAs [7,10]. As fore-mentioned phase evolution, at high temperature (1000 °C), reordering could result in more stable phases during the SPS process. Thus, the formation of the FCC phase is mainly attributed to the reordering during SPS at high temperature and high mixing entropy of FeNiCrCo_{0.3}Al_{0.7} HEA system.

It can be concluded that the phases of FeNiCrCo_{0.3}Al_{0.7} HEA consists of a disordered BCC phase (Fe–Cr-rich phase), an ordered BCC (Ni–Al-rich phase) and an FCC phase. It is impossible that all elements are expected to be randomly distributed in the crystal lattice in HEAs. Though HEAs systems exhibit high mixing entropy, some ordered phases which are deviated from the definition of HEAs can also be formed. It reveals that high mixing entropy is insufficient to dominate the formation of phases in HEAs systems, the enthalpy of mixing also can play a significant role of the formation of HEAs phases [17,36].

3.3. Mechanical properties

Bulk FeNiCrCo_{0.3}Al_{0.7} HEA after consolidation of SPS possesses excellent mechanical properties. Fig. 5 shows the stress–strain curve of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS under compression at room temperature. The yield strength (σ_y), compressive strength (σ_{max}) and compression ratio (ε_p) of FeNiCrCo_{0.3}Al_{0.7} HEA are 2033 ± 41 MPa, 2635 ± 55 MPa and $8.12 \pm 0.51\%$, respectively. The average Vickers hardness of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS has been measured to be $624 \pm 26 H_v$. Moreover, plastic deformation behavior is also distinctly characterized in the stress–strain curve, indicating the alloy shows limited plastic property. The mechanical properties of FeNiCrCo_{0.3}Al_{0.7} HEA and some typical HEAs of FeNiCrCoAl HEA system are listed elaborately in Table 4. The listed HEAs are processed by casting (arc-melting) except FeNiCrCo_{0.3}Al_{0.7} HEA studied in the present work. Compared with FeNiCrCoAl HEA, the yield strength of FeNiCrCo_{0.3}Al_{0.7} HEA has increased by 62.5% with an increment of compressive strength by 31.5%. However, compression ratio of FeNiCrCo_{0.3}Al_{0.7} HEA is much lower than

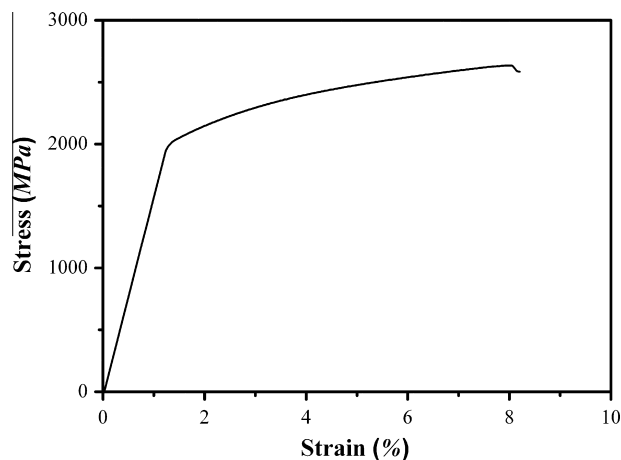


Fig. 5. Stress–strain curve of bulk FeNiCrCo_{0.3}Al_{0.7} HEA under compression.

that of FeNiCrCoAl HEA. FeNiCrCo_{0.3}Al_{0.7} HEA exhibits the highest Vickers hardness in comparison with typical HEAs of FeNiCrCoAl HEA system listed in Table 4. Besides, according to all reported works, the compressive yield strength, fracture strength and compression ratio of mostly investigated HEAs are not more than 1800 MPa, 2200 MPa and 10%, respectively [28]. Apparently, FeNiCrCo_{0.3}Al_{0.7} HEA still shows better compressive properties compared with a lot of typical HEAs which have been reported.

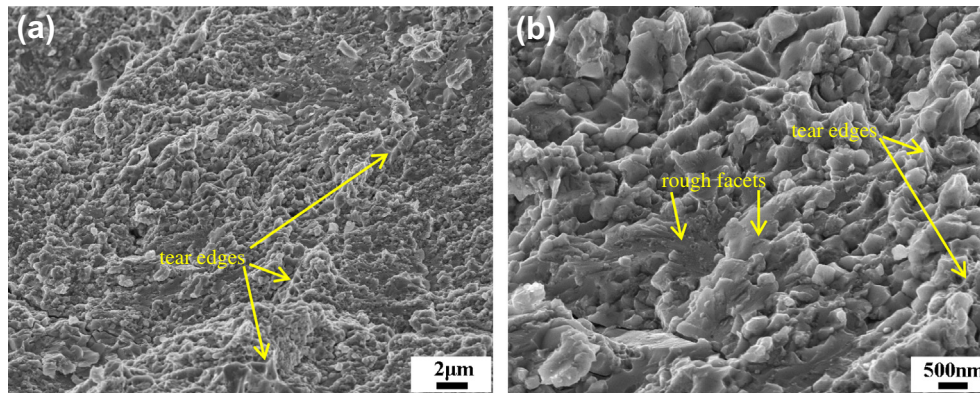
The above-discussed results confirm that FeNiCrCoAl HEA may possess appropriate concentrations of all component elements. Furthermore, compared with other typical HEAs of FeNiCrCoAl HEA system listed in Table 4, FeNiCrCo_{0.3}Al_{0.7} HEA exhibits distinctly much better mechanical properties except compression ratio. As discussed proceeding, the decrement of the concentration of Co equals to the increment of the concentration of Al in FeNiCrCo_{0.3}Al_{0.7} HEA (as shown in Table 2) can strengthen HEAs due to improving the effect of solid solution strengthening [38]. BCC phase often displays higher strength and lower plasticity than FCC phase, and there are two BCC structure phases in FeNiCrCo_{0.3}Al_{0.7} HEA, especially the BCC2 phase (Ni–Al-rich phase) displaying a B2 structure is a typical high-strength phase. Therefore, FeNiCrCo_{0.3}Al_{0.7} HEA has higher yield strength and compressive strength compared with FeNiCrCoAl HEA. In addition, FeNiCrCo_{0.3}Al_{0.7} HEA prepared by MA and SPS displays ultrafine grains, which can also strengthen the alloy. In conclusion, FeNiCrCo_{0.3}Al_{0.7} HEA exhibits the highest Vickers hardness owing to solid solution strengthening mechanism of Al atoms and the two BCC phases, as well as ultrafine grains. Furthermore, the limited compression ratio of FeNiCrCo_{0.3}Al_{0.7} HEA is mainly attributed to the porosity and the two brittle BCC phases, especially the BCC2 phase with B2 structure. It is worth mentioning that the plasticity of the alloy is mainly contributed by the FCC phase.

Fracture images for bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS are shown in Fig. 6. As three phases coexist in the structure, the small binding force among these three phases facilitates to the compression ratio (ε_p) of only $8.12 \pm 0.51\%$. Fig. 6a shows low-magnified fracture image, short tear edges are observed clearly. In order to investigate the fracture mechanism of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS, high-magnified fracture image is presented in Fig. 6b. A few short tear edges, rough facets and typically intercrystalline fracture feature are clearly characterized, demonstrating the fracture form of the alloy is typically intercrystalline fracture accompanied by quasi-cleavage fracture. Obviously, the short tear edges are generated from the deformation of the FCC phase, which has beneficial effects on the plasticity. However, the amount of short tear

Table 4

Mechanical properties of HEAs at room temperature.

Alloy	Process	σ_y (MPa)	$\sigma_{\max}$ (MPa)	ε_p (%)	Hardness (H_v)	Refs.
FeNiCrCo _{0.3} Al _{0.7}	SPS	2033 ± 41	2635 ± 55	8.12 ± 0.51	624 ± 26	This work
FeNiCrCoAl	Arc-melting	1251	2004	32.7	–	[21]
FeNiCrCoAl _{0.25}	Arc-melting	–	–	–	110 ± 2	[23]
FeNiCrCoAl _{0.5}	Arc-melting	–	–	–	159 ± 2	[23]
FeNiCrCoAl _{0.75}	Arc-melting	–	–	–	388 ± 5	[23]
FeNiCrCoAl _{0.875}	Arc-melting	–	–	–	538 ± 12	[23]
FeNiCrCoAl ₁	Arc-melting	–	–	–	484 ± 26	[23]
FeNiCrCoAl ₂	Arc-melting	–	–	–	509 ± 27	[23]

**Fig. 6.** Fracture images of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS. (a) Low-magnified image and (b) high-magnified image.

edges is fairly small, implying limited plasticity. The rough facets of Fig. 6b might correspond to the deformation of Region 1 of Fig. 3b. Moreover, the two BCC phases with high strength and low plasticity might yield the intercrystalline fracture. According to the proceeding, the two BCC phases are the main phases. Thus, FeNiCrCo_{0.3}Al_{0.7} HEA after SPS exhibits high strength and the fracture form of the alloy is typically intercrystalline fracture accompanied by quasi-cleavage fracture. In conclusion, the main fracture mechanism is a mixture of intercrystalline fracture and quasi-cleavage fracture, which is consistent with the limited compression ratio ($8.12 \pm 0.51\%$) of FeNiCrCo_{0.3}Al_{0.7} HEA after SPS.

4. Conclusions

FeNiCrCo_{0.3}Al_{0.7} HEA has been successfully designed and synthesized. During MA, all of other elements appear to be dissolving into Fe or Cr peaks as the milling time increases, and finally a supersaturated solid solution with BCC structure was formed. After SPS, an FCC phase and a BCC phase are observed in the XRD pattern. The SEM micrographs show that the phases of FeNiCrCo_{0.3}Al_{0.7} HEA exhibit three kinds of distinctive morphologies. The TEM analysis results suggest that FeNiCrCo_{0.3}Al_{0.7} HEA consists of two BCC phases having the same lattice parameter and one FCC phase. The BCC1 phase with A2 structure is enriched in Fe–Cr and the BCC2 phase with B2 structure is enriched in Ni–Al, these two BCC phases are evidently deviated from the definition of HEAs, whereas the FCC phase responds to the definition. The stoichiometry of these two BCC phases could be written as $(\text{Ni}_x\text{Co}_y\text{Cr}_z)_{1-x-y-z}(\text{Al}_v\text{Fe}_w)$, but their chemical compositions are on different levels. The yield strength, compressive strength, compression ratio and Vickers hardness of FeNiCrCo_{0.3}Al_{0.7} HEA are 2033 ± 41 MPa, 2635 ± 55 MPa, $8.12 \pm 0.51\%$ and $624 \pm 26H_v$, respectively. FeNiCrCo_{0.3}Al_{0.7} HEA exhibits distinctly much better mechanical properties except compression ratio compared with other typical HEAs of FeNiCrCoAl HEA system. The main fracture mechanism of bulk FeNiCrCo_{0.3}Al_{0.7} HEA after SPS is intercrystalline fracture and quasi-cleavage fracture.

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